LLMWiki

We saved $400/month on day one by NOT using the obvious choice.

The full LLMWiki architecture on AWS

and every decision behind it.

S3 → Textract → Bedrock → S3 Vectors → Lambda

■ Distributed ■ Fault-tolerant ■ Secured ■ Scalable ■ Validated

Swipe → full architecture breakdown, every decision explained

THE PROBLEM

This is how most enterprises manage critical knowledge today.

- Critical docs buried in shared drives nobody searches
- Tribal memory — one person holds the answer
- Email thread to find the expert — 2 days wait
- Answer walks out the door when they resign

Sound familiar? Swipe to see the fix →

THE INGESTION PIPELINE

Any format. Any size.
3 minutes to queryable.

PDF / Office → **Scanned Docs** → **Runbooks**

Amazon S3 ← EventBridge trigger on upload

AWS Textract → **Amazon Bedrock** → **S3 Vectors**

PDF → Markdown

Chunk + Embed

Vector Store■~$0/mo base

✓ Pure event-driven. Zero idle cost.

No polling. No scheduled jobs. Fires on every upload.

THE QUERY PIPELINE

Question in. Cited answer out.

User Question → **Lambda** → **Bedrock KB** → **Claude** → **Cited Answer**

Confidence Score attached to every answer:

HIGH
2+ direct-match sources

MEDIUM
Partial match sources

LOW
Gap Detection triggered

Every answer includes the source wiki pages that were used.

No hallucination without citation. Gaps are flagged, not fabricated.

RAG retrieval + Claude synthesis in one Lambda invocation

THE DATA LAYER

Every query logged.
Every source tracked.

llmwiki-index
Page registry — every wiki document, · its slug, type, tags, confidence baseline

llmwiki-log
Full audit trail — every query, every · agent invocation, every confidence score

llmwiki-source-registry
Document provenance — uploader, timestamp, · trust level, KB sync status

Compliance-ready audit trail out of the box.

DynamoDB — serverless, scales to zero, pay per request

THE COST DECISION

Same capability.
1/∞ the floor cost.

OpenSearch

$700+/month

minimum — whether
you use it or not

Always-on capacity
OCU billing
Complex IAM setup

VS

■ S3 Vectors

~$0/month base

pay per query
zero idle cost

Serverless native
S3 IAM — already set up
Same vector search

We chose S3 Vectors on day one. Still the right call.

THE FIVE PILLARS

Production, not a demo.

- **DISTRIBUTED** — Event-driven ingestion + retrieval
  Independent components — no bottleneck
- **FAULT-TOLERANT** — Idempotent Lambdas · DLQs · DynamoDB state
  Automatic retry — no manual intervention
- **SECURED** — KMS encryption at rest + in transit
  IAM least-privilege per Lambda · VPC isolation
- **SCALABLE** — Serverless-first — scales to zero
  Scales to millions of documents · no rearchitecting
- **VALIDATED** — 22 unit tests generated by Codex
  First-pass: 22 passed · 0 failed · CI-ready

LLMWiki runs in production — not as a proof-of-concept

WHAT'S NEXT

This architecture just got a brain.

Next week: Neuro-SAN & the AAOSA Protocol

5 specialist agents collaborating on one customer question.
No agent ever sees the sensitive data.
Orchestrated in HOCON — no Python business logic required.

Want to see this on your own documents?
30 minutes · Live demo · DM me on LinkedIn